EART11300

Igneous, sedimentary and metamorphic rocks in hand specimen

Video 3: Attributes of **metamorphic** rocks

Metamorphic rocks

When you are looking at a hand specimen or thin section:

- Compositional layering (fossil bedding) may be present, but may be deformed (folded, etc).
- Contain relatively few minerals (normally 6 or less); crystals of each mineral tend to be quite uniform.
- Crystals often have a preferred orientation (tectonic fabric) produced by deformation. If the fabric is planar (foliation), it may be discordant to any layering present.
- Some crystals have their own shape; others have shapes indicative of competitive growth in the solid state (analogy: bubbles in foam).
- No significant porosity.
- No fossils usually survive.

Metamorphic rocks

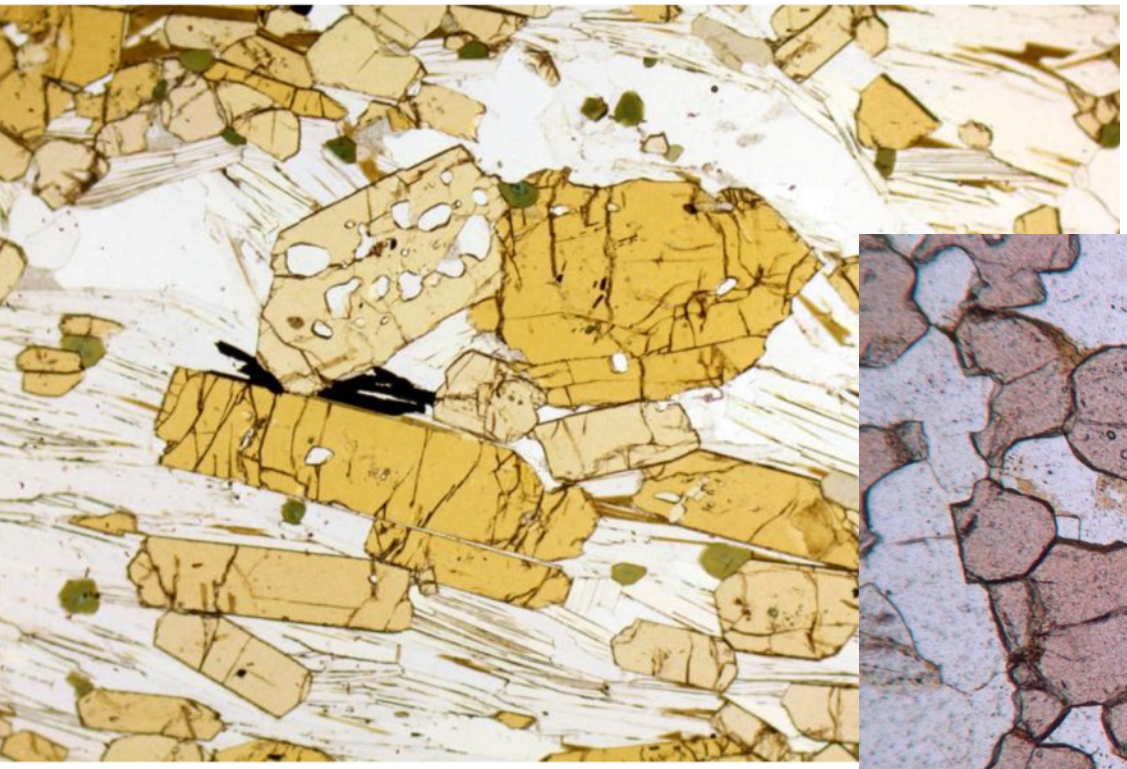
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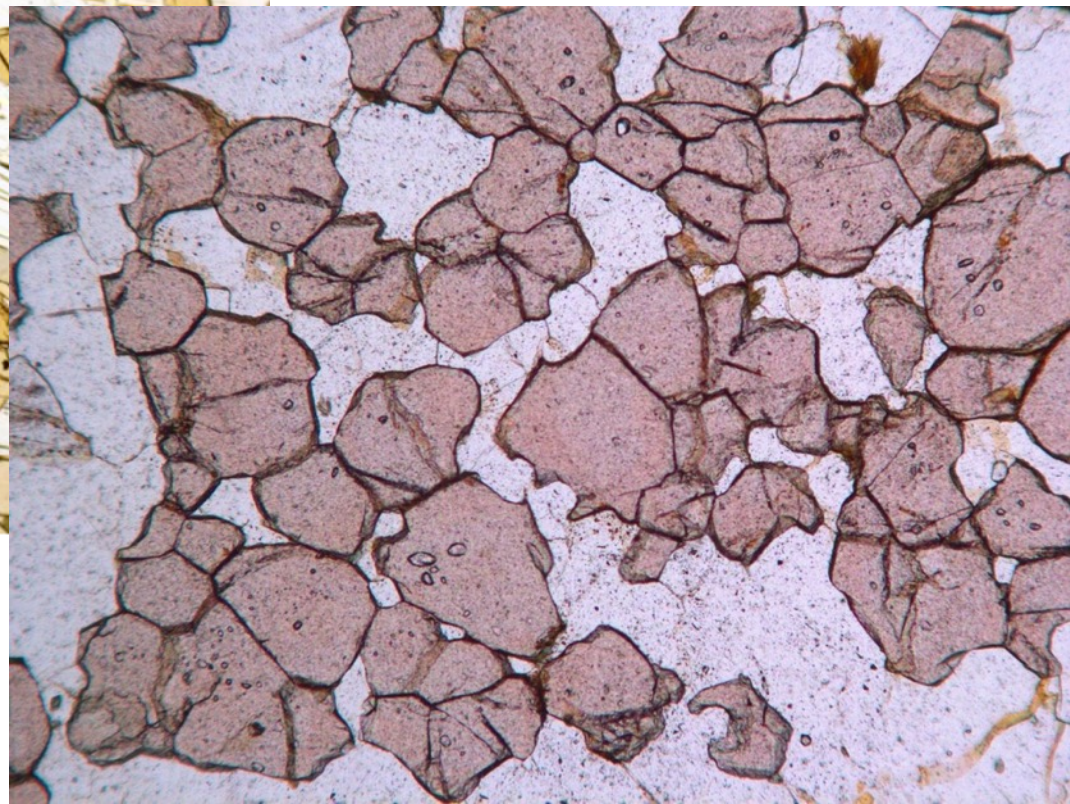




Metamorphic rocks

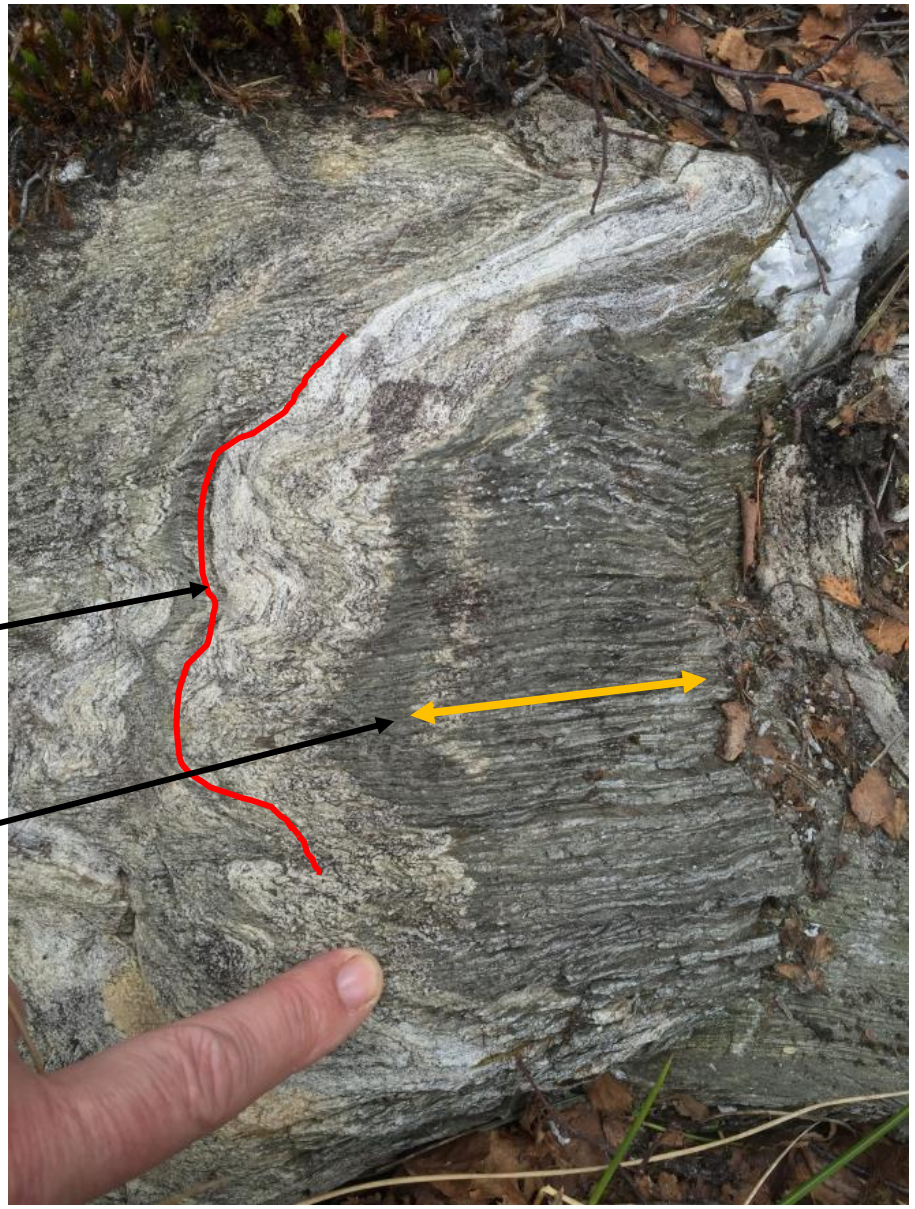
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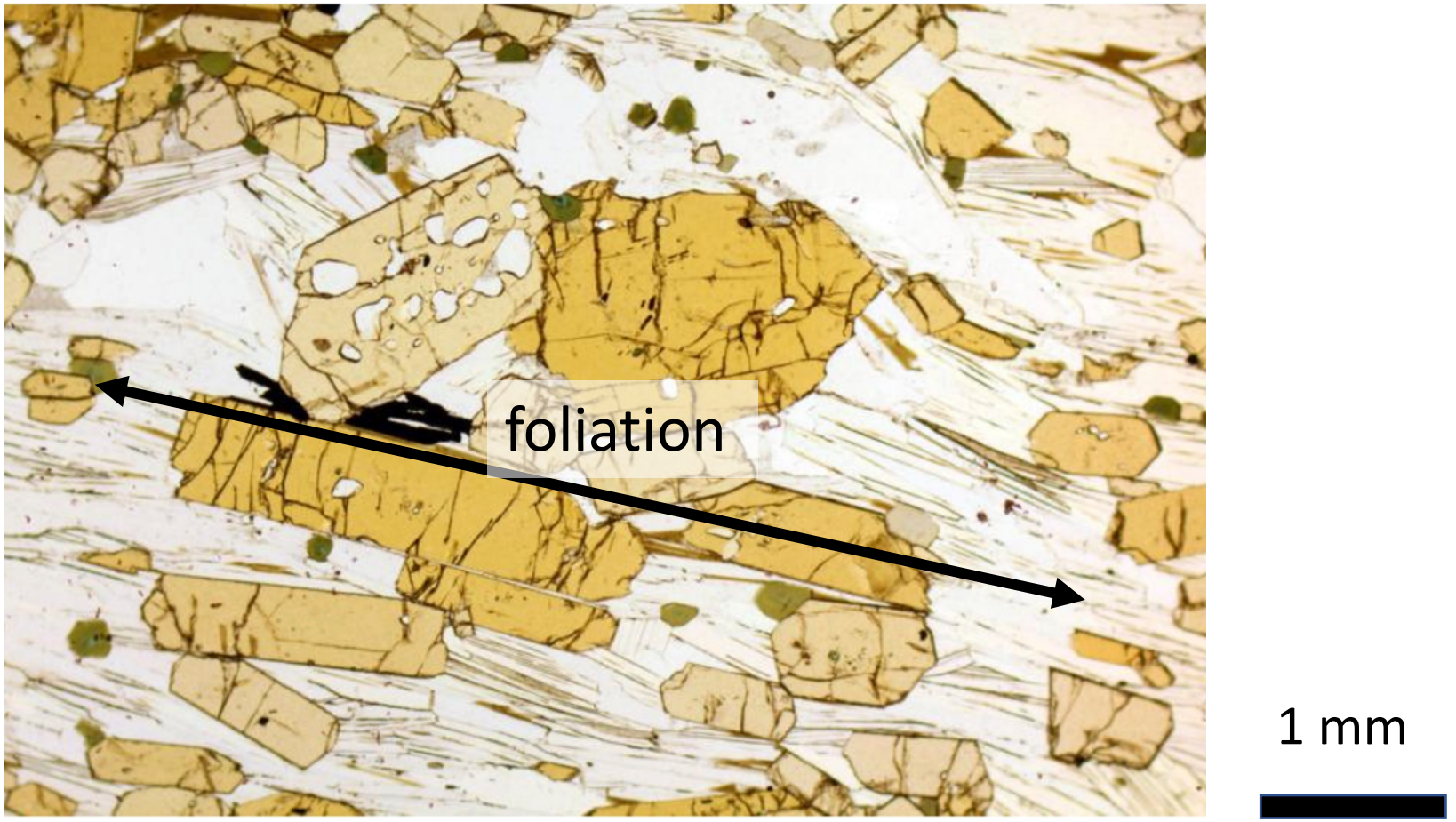
Metamorphic rocks

- Crystals often have a preferred orientation (tectonic fabric) produced by deformation. If the fabric is planar (foliation), it may be discordant to any layering present.

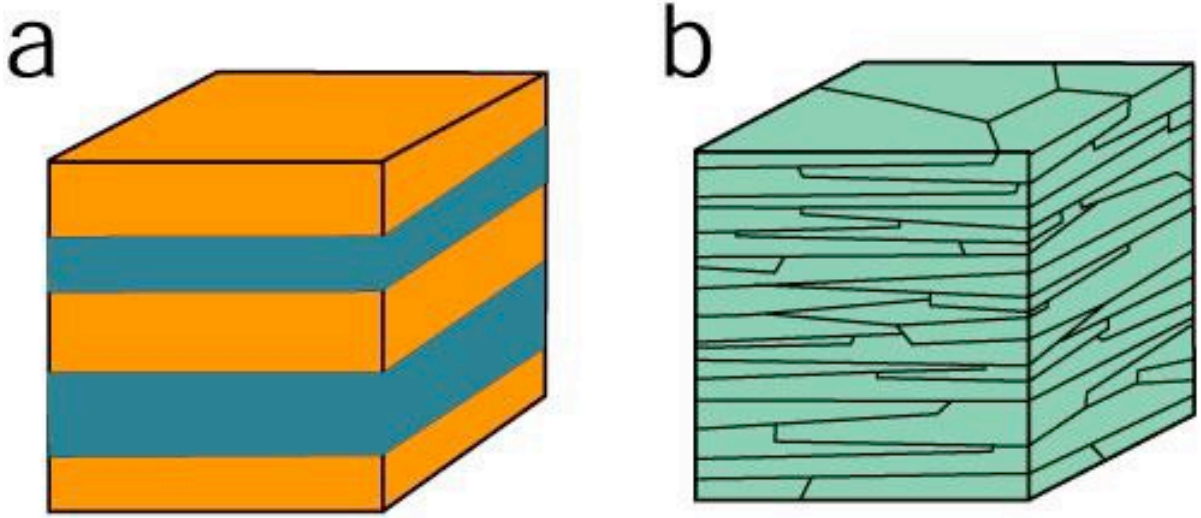


Metamorphic rocks

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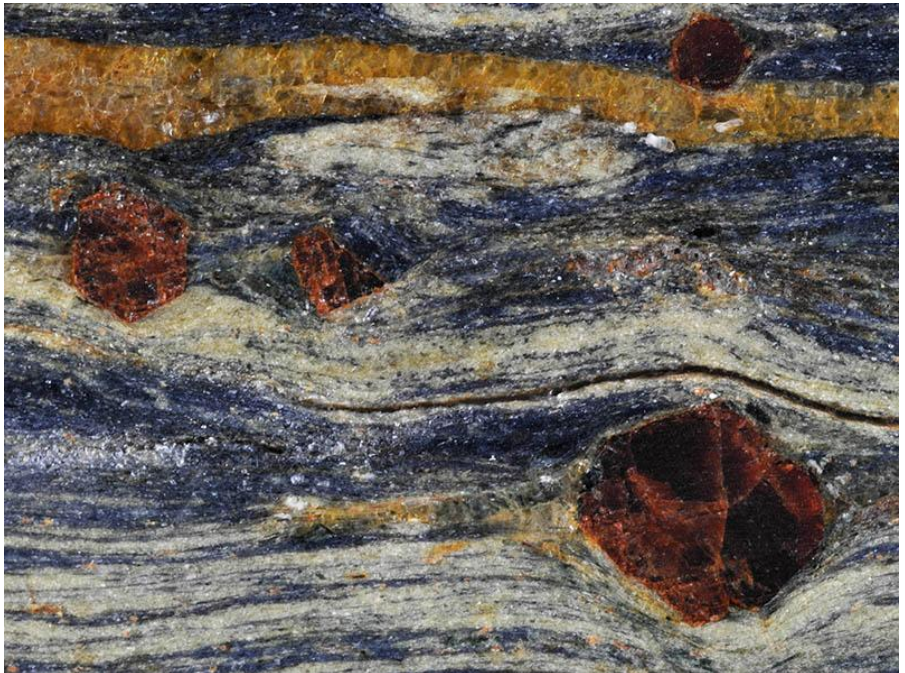
Difference between (a) compositional layering and (b) foliation:

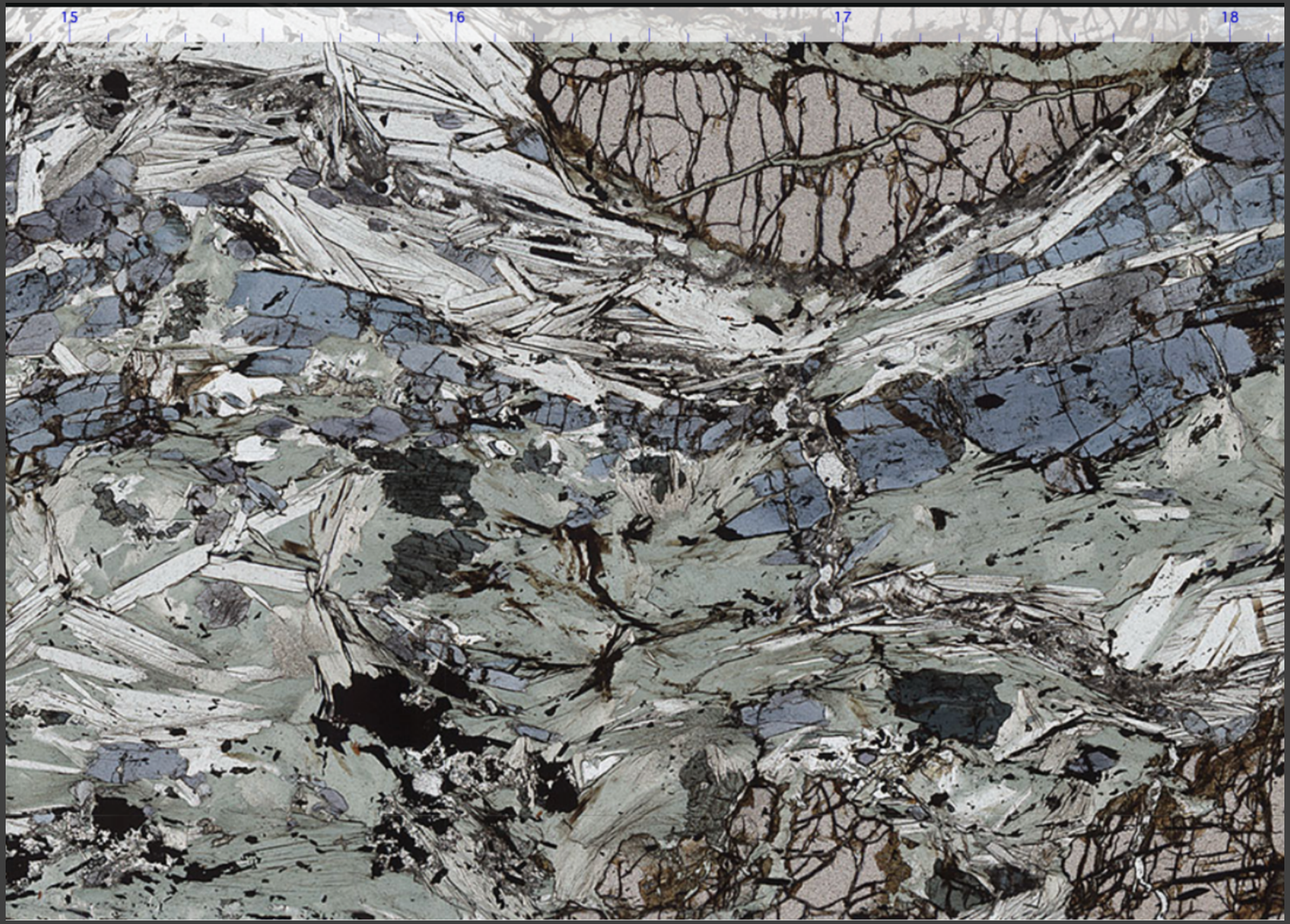


a. Layers have different compositions

b. Flaky minerals are all in the same orientation

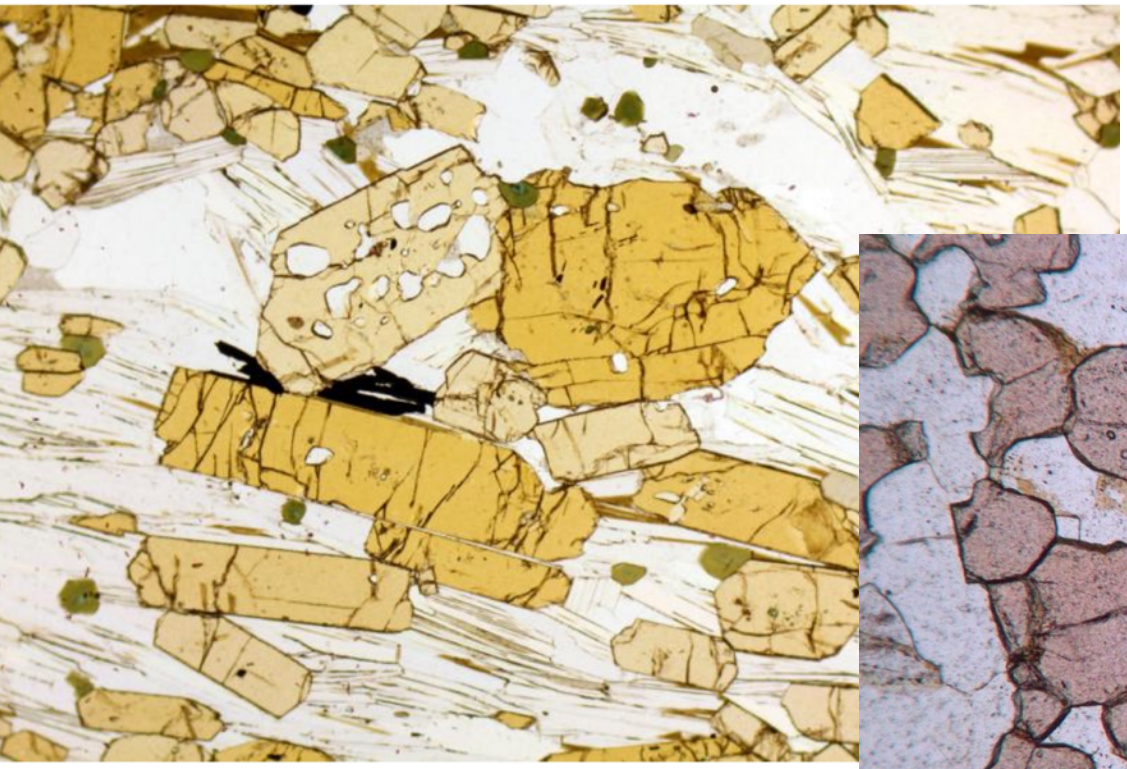


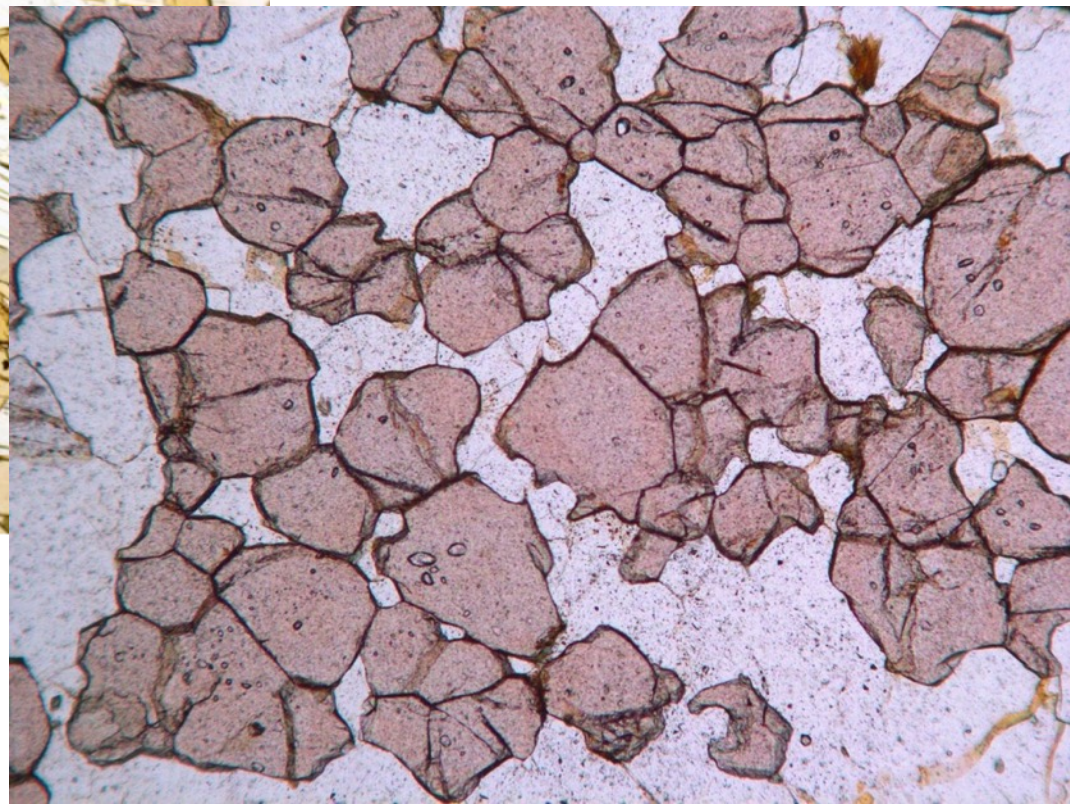



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Metamorphic rocks

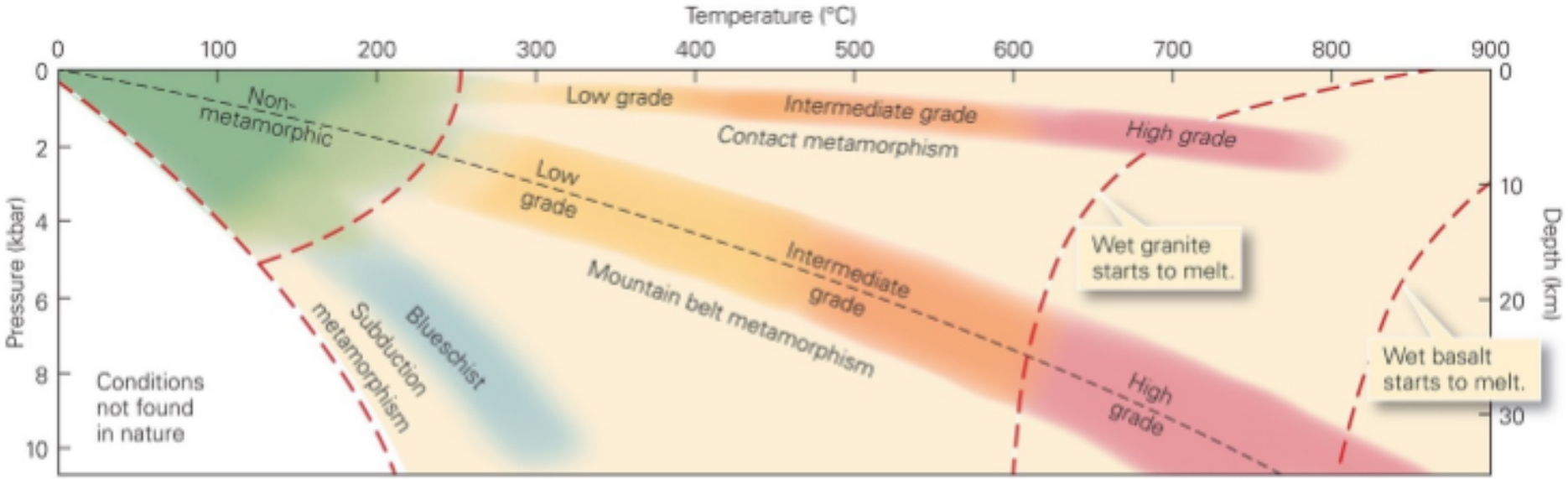
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Metamorphic rocks

- No significant porosity.

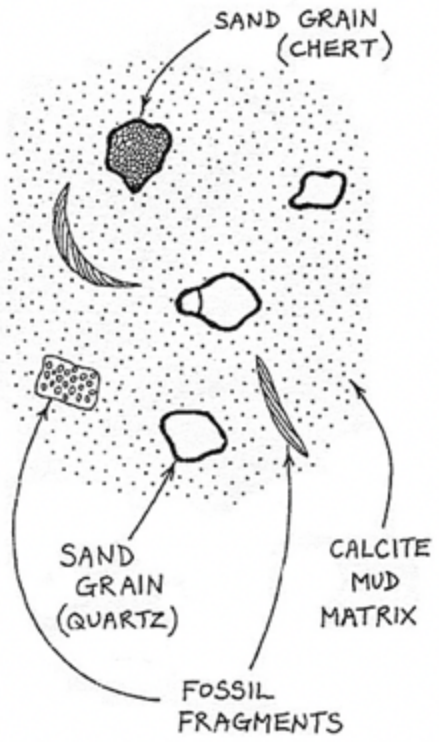


(a) This graph depicts the approximate temperatures and pressures of metamorphic grades. Different conditions occur in different geologic settings.

Metamorphic rocks

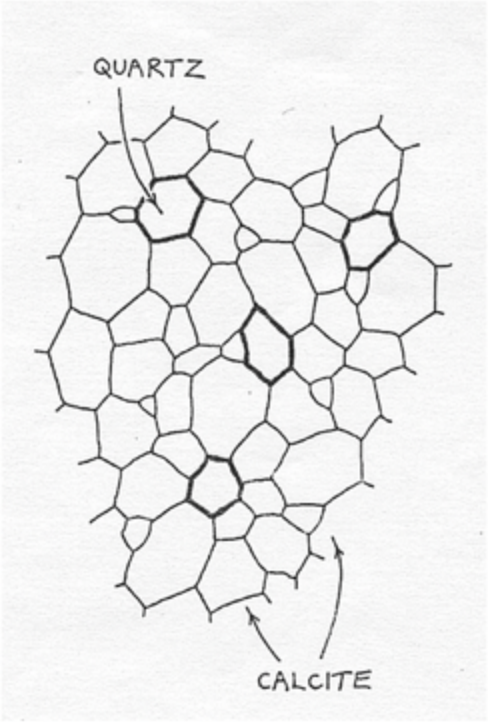
- No fossils usually survive.

Initial rock:



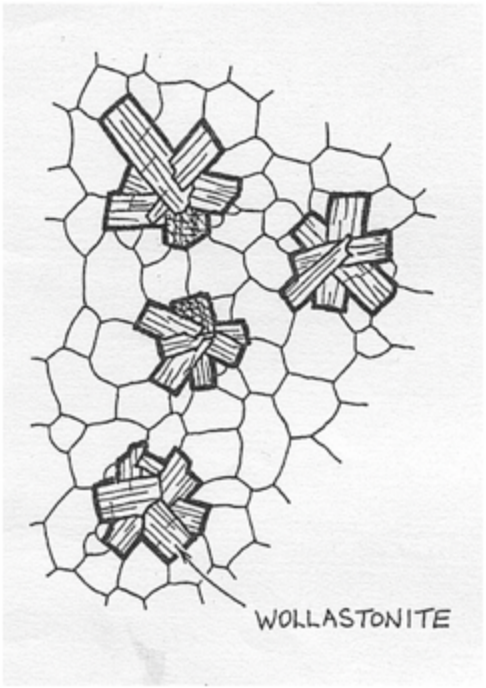
Recrystallisation

After significant T increase:



Further recrystallisation & reaction

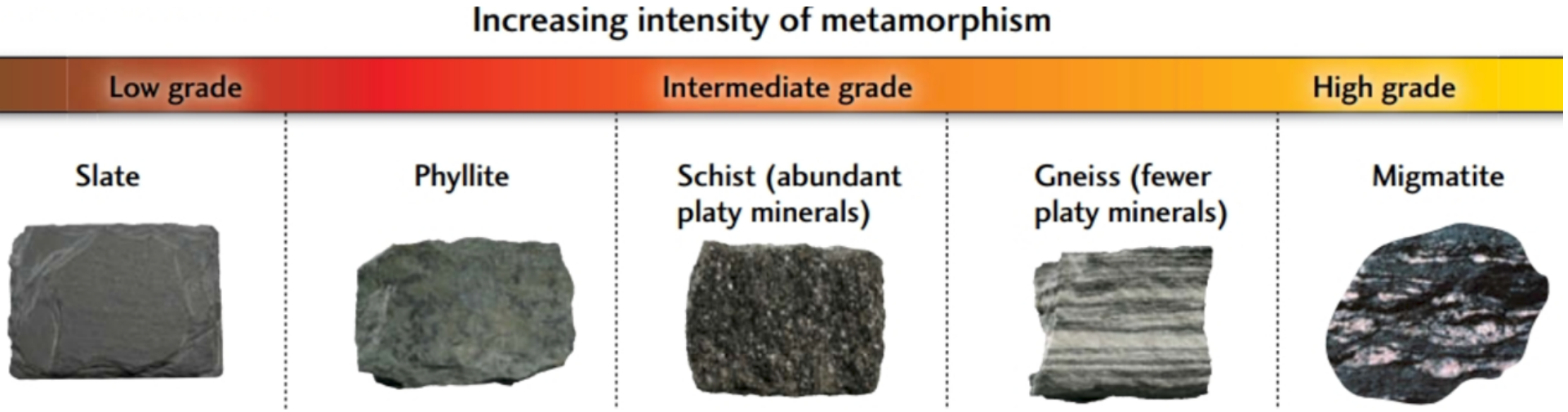
After further T increase:



Further classification of metamorphic rocks

Metamorphic rocks can be classified according to the extent that the rock has changed during metamorphism, which usually depends on the temperature reached, and is known as **metamorphic grade.**

- **Low grade** display the least amount of change. If the rock started out as fine-grained (e.g. a mudstone), then low-grade metamorphism will result in a rock that is still fine-grained.
- **Medium grade** show an intermediate level of change. It will be coarser grained.
- **High grade** show the most change. It will have the largest crystals.



Grotzinger & Jordan Fig. 6.6